INCH POUND

MIL-PRF-52345D
19 March 1999
SUPERSEDING
MIL-F-52345C (ME)
4 August 1989

PERFORMANCE SPECIFICATION

FLOW CONTROL KIT, HOSELINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Abstract. This specification covers a hoseline flow control kit for the distribution of bulk liquid fuel with a 4-inch collapsible hoseline.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specification, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-D/210, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3835

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SPECIFICATIONS

FEDERAL

A-A-52598 – Couplings, Clamp, Pipe; with Bolts and Synthetic Rubber Gaskets for Grooved-End Pipe
A-A-59326 – Coupling Halves, Quick-Disconnect Cam-Locking Type
A-A-59379 – Valves, Gate, Rising Stem, Double Acting, Lightweight

DEPARTMENT OF DEFENSE

MIL-PRF-370 – Hose and Hose Assemblies, Nonmetallic: Elastomeric, Liquid Fuel

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available, from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents that are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C606 – Grooved and Shouldered Joints (DoD adopted)

(Application for copies should be addressed American Water Works Association, 6666 West Quincy Avenue, Denver, Colorado 80235.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The flow control kit is intended for use in controlling the fuel flow in a lightweight, collapsible, 4-inch hoseline system. Components shall be compatible with a wide range of military fuels, to include JP-8, automotive gasolines and diesel. The kit shall be contained in a container to allow ease of transport and control of components.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Kit components. The flow control kit shall include the components listed in table I. The components of the kit, in any appropriate coupling, shall withstand a hydrostatic pressure of 150 pounds per square inch (psi), minimum, without leakage, deformation, fracture, malfunction, or degradation of overall performance.

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TABLE I. Flow control kit components' list.

Item	Component	Ref.	Quantity
1	Gate valves	3.3.1	2
2	Tee	3.3.2	1
3	Pipe coupling clamp	3.3.3	8
4	Check valves	3.3.4	2
5	Hose assembly	3.3.5	2
6	Strainer assembly	3.3.6	1
7	Grooved-to-external thread coupling	3.3.7	2
8	Grooved-to-internal thread coupling	3.3.8	2
9	Male coupling half	3.3.9	2
10	Female coupling half	3.3.10	2
11	Fish tape	3.3.11	1

3.3.1 Gate valves. The gate valves shall be a 4-inch, rising stem in accordance with A-A-59379, Type II, except each end of the valve shall be a 4-inch, single grooved end, in accordance with AWWA C606-97, Table 4 (Cut grooving dimensions - Steel, aluminum, brass, and other metallic pipe of IPS dimensions).

3.3.2 Tee. The tee shall be aluminum, 4-inch, and grooved at all ends in accordance with AWWA C606-97, Table 4 (Cut grooving dimensions - Steel, aluminum, brass, and other metallic pipe of IPS dimensions).

3.3.3 Pipe coupling clamp. The pipe coupling clamps shall be aluminum, 4-inch, in accordance with A-A-52598.

3.3.4 Check valves. The check valves shall be 4-inch certifiable for a working pressure of 300 pound per square inch (psi). The ends shall be grooved in accordance with AWWA C606-97, Table 4 (Cut grooving dimensions - Steel, aluminum, brass, and other metallic pipe of IPS dimensions).

3.3.5 Hose assembly. The hose assembly shall consist of a 60.00-inch long section of 4-inch, high pressure, non-reinforced, collapsible, lightweight elastomeric tube in accordance with MIL-PRF-370, with hose coupling adapters secured at each end. The hose adapters shall be single grooved end, 4-inch I.D., 10.00 inches long, in accordance with AWWA C606-97, Table 4 (Cut grooving dimensions - Steel, aluminum, brass, and other metallic pipe of IPS dimensions).

3.3.6 Strainer assembly. The strainer assembly shall accept a strainer configuration like that shown in figure 1, housed in a hose coupling adapter, grooved both ends, 4-inch I.D., 10.00-inches long, in accordance with AWWA C606-97, Table 4 (Cut grooving dimensions - Steel, aluminum, brass, and other metallic pipe of IPS dimensions). The strainer shall remove foreign material .125 ±. .005 inch in diameter and larger. Materials incorporated into the assembly shall be compatible with all components of the flow control kit. The materials incorporated shall be capable of meeting all the operational and environmental requirements specified and be reusable without degradation of performance.

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3.3.7 Grooved-to-external thread coupling. The coupling shall be a 4-inch fitting in accordance with AWWA C606-97, Table 4 (Cut grooving dimensions - Steel, aluminum, brass, and other metallic pipe of IPS dimensions) having one end grooved and the other having external pipe threads. The coupling length shall not exceed 6.00 inches.

3.3.8 Grooved-to-internal thread coupling. The coupling shall be a 4-inch fitting in accordance with AWWA C606-97, Table 4 (Cut grooving dimensions - Steel, aluminum, brass, and other metallic pipe of IPS dimensions) with one end grooved and the other end having internal pipe threads.

3.3.9 Male coupling half. The male coupling shall be a 4-inch, male cam-locking coupling half in accordance with A-A-59326, with the other end having an external pipe thread.

3.3.10 Female coupling half. The female coupling shall be a 4-inch, female cam-locking coupling half in accordance with A-A-59326, with the other end having an internal pipe thread.

3.3.11 Fish tape. The tape shall be the type used by electricians to pull wires through conduit or other closed areas. The tape shall be 100 feet in length or greater, on a reel with a winder.

3.3.12 Container. A container shall be provided for the components of the kit. The container shall not be greater than 22 inches in height. The container shall be sufficiently strong to support all components in rough handling. The container shall have bins, cradles, or blocking to prevent component damage during shipment and handling. The container shall have a hinged cover and latches to secure the cover in the closed position. The cover shall be self-supporting in the open position. A loading diagram shall be fixed to the inside of the lid, visible to the operator while loading. The diagram shall provide pictorial representation for loading kit components into the container. Based on a 40-pound lift capability per handle, handles shall be provided in sufficient number to permit lifting by personnel. The container shall have provisions for mechanical lifting by forklift and sling lifting.

3.4 Support and ownership requirements.

3.4.1 Materials. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The materials shall be of sufficient durability to meet all the performance requirements as specified herein. The material shall have no adverse effect on the health of the personnel when used for its intended purpose. Whenever feasible, toxic chemicals, hazardous substances, or ozone depleting chemicals (ODC) shall not be used.

3.4.1.1 Deterioration prevention and control. The equipment shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration to which the materials are susceptible.

3.4.1.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion (see 6.6.1).

3.4.1.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes

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economically advantageous life cycle costs. Used, rebuilt or re-manufactured parts shall not be used in the repair kit (see 6.6.2).

3.4.2 Color and marking. Color of the container and components shall be as specified by the contract or order (see 6.2). Markings of the container and all components shall be in accordance with contract or order instructions (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article inspection shall be performed on a complete flow control kit(s) (see 3.2). This inspection shall include the examination of 4.4 and the tests of 4.5. Nonconformance to any specified requirement, the failure of any test, or the presence of one or more defects shall be cause for rejection.

4.3 Conformance inspection. Conformance inspection shall be performed on all flow control kit production units and shall include the applicable examinations of 4.4. Nonconformance to any specified requirement or the presence of one or more defects shall be cause for rejection.

4.4 Examination. The flow control kit components shall be examined for defects as specified in table II.

TABLE II. Examination schedule.

Exam	First article	Conf.	Description	Reference paragraph	Method of inspection
101	X	X	Kit component quantities not as specified.	3.3	Visual
102	X	X	Gate valves not as specified.	3.3.1	Visual/Doc
103	X	X	Tee not as specified.	3.3.2	Visual
104	X	X	Pipe coupling clamps not as specified.	3.3.3	Visual
105	X	X	Check valves not as specified.	3.3.4	Visual
106	X	X	Hose assemblies not as specified.	3.3.5	Visual
107	X	X	Strainer assembly not as specified.	3.3.6	Visual/Doc
108	X	X	Grooved-to-external thread coupling not as specified.	3.3.7	Visual
109	X	X	Grooved to-internal thread coupling not as specified.	3.3.8	Visual
110	X	X	Male coupling half, external pipe thread, not as specified.	3.3.9	Visual

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TABLE II. Examination schedule – Continued

111	X	X	Female coupling half, internal pipe thread, not as specified.	3.3.10	Visual
112	X	X	Fish tape not as specified.	3.3.11	Visual
113	X	X	Container not a specified.	3.3.12	Visual/SIE
114	X	X	Material not as specified.	3.4.1	Visual/Doc
115	X	X	Color and markings not as specified.	3.4.2	Visual

Doc = Documentation; SIE = Standard Inspection Equipment

4.5 Tests.

4.5.1 Test conditions. Testing shall be conducted at ambient temperatures between 35 degrees Fahrenheit (°F) and 120°F. Water shall be used as the test medium. The testing shall require a means of maintaining a pressure of 150 psi on the water filled assembly for a period of at least five minutes. Failure to maintain the required pressure for the designated time shall constitute reason for retesting.

4.5.2 Component testing. The component test shall be conducted after the lift test using the same kit that was subjected to the lift test. From the container components, assemble the following test system. Take one 4-inch, female, quick-disconnect coupling half and one grooved to external threaded coupling, and attach the external threaded coupling to the quick disconnect coupling half. Using pipe coupling clamps, attach one hose assembly to the grooved end of the external threaded coupling, and attach the tee to the other hose end. With a second hose assembly and pipe coupling clamps, attach one end of the hose to one leg of the tee and the other end of the hose to a 4-inch gate valve. To the remaining tee leg, using a pipe coupling clamp, attach a check valve. The check valve shall be connected to stop the flow. Through the quick-disconnect coupling half, fill the assembly with water to a pressure of 150 psi, making certain all air is removed from the system, and close the gate valve. Immediately check the assembly for leaks and pressure loss. If there are no leaks or pressure loss, maintain the 150 psi pressure for a period of not less than five minutes. During the time the assembly is under pressure check for leaks and other defects. At the end of the required time, slowly lower the pressure to ambient and recover the components. Any damage to components, leaks, or loss of pressure during the five minute period shall constitute failure of this test.

4.5.3 Container test. The container, with contained components, shall be lifted by mechanical means utilizing lift provisions. The container shall be lifted not less than 48 inches off the ground and the lift shall be repeated five times with at least one lift at the maximum speed of the lifting equipment. At the conclusion of lift five, the container shall be examined for damage, or deterioration. Open the container and examine the interior and component for mislocation and other damage. Failure of the container to maintain compartmental and structural integrity through all lifts shall constitute failure of this test.

4.5.4 Strainer test. The strainer shall be subjected to a 350 gallon per minute flow of liquid containing solid foreign material between 0.06 and 0.25 inches in diameter for a period of not less than five minutes. The ratio of solids to liquid shall not be less than one part per hundred by

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volume. The foreign matter effluent from the strainer shall be checked for size. Failure of the strainer to remove foreign matter 0.125-inches and larger shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature, which may be helpful but is not mandatory.)

6.1 Intended use. The flow control kit is a military unique system for operation in an extreme environment, under conditions typically found on the battlefield. The specific components as a kit in a container are not available commercially and must interact without fault from material or compatibility failures. The components of the flow control kit are intended for use in directing the flow in a bulk liquid fuel distribution system; to interconnect facilities for the receipt, storage, and dispensing of a wide range of military fuels. The kit shall be a component of the 4-inch hoseline outfit.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this publication.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see Section 2).
c. When a first article is required for inspection and approval, time frame for submission, and the number of units required (see 3.2).
d. Color and markings required (see 3.6).
e. Packaging required (see 5.1).

6.3 First article. When a first article inspection is required, the item(s) should be a first article production unit. The contracting officer should include specific instructions in acquisitions' documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first article(s).

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including special tools), and instructions on shipment of the flow control kit.

6.5 Data requirements. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and

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standards that have been cleared and listed in DoD 5010.12 L, Acquisition Management Systems and Data Requirements Control List (AMSDL) should be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals should be acquired under a separate contract line item in the contract.

6.6 Definitions.

6.6.1 Dissimilar metals. For its purpose, this document considers metals dissimilar when two specimens in contact with each other promote accelerated galvanic corrosion (see 3.4.1.2).

6.6.2 Recovered materials. For the purpose of this document, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials (see 3.4.1.3).

6.7 Subject term (key word) listing.

Couplings
Distribution
Liquid fuel
Valve

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

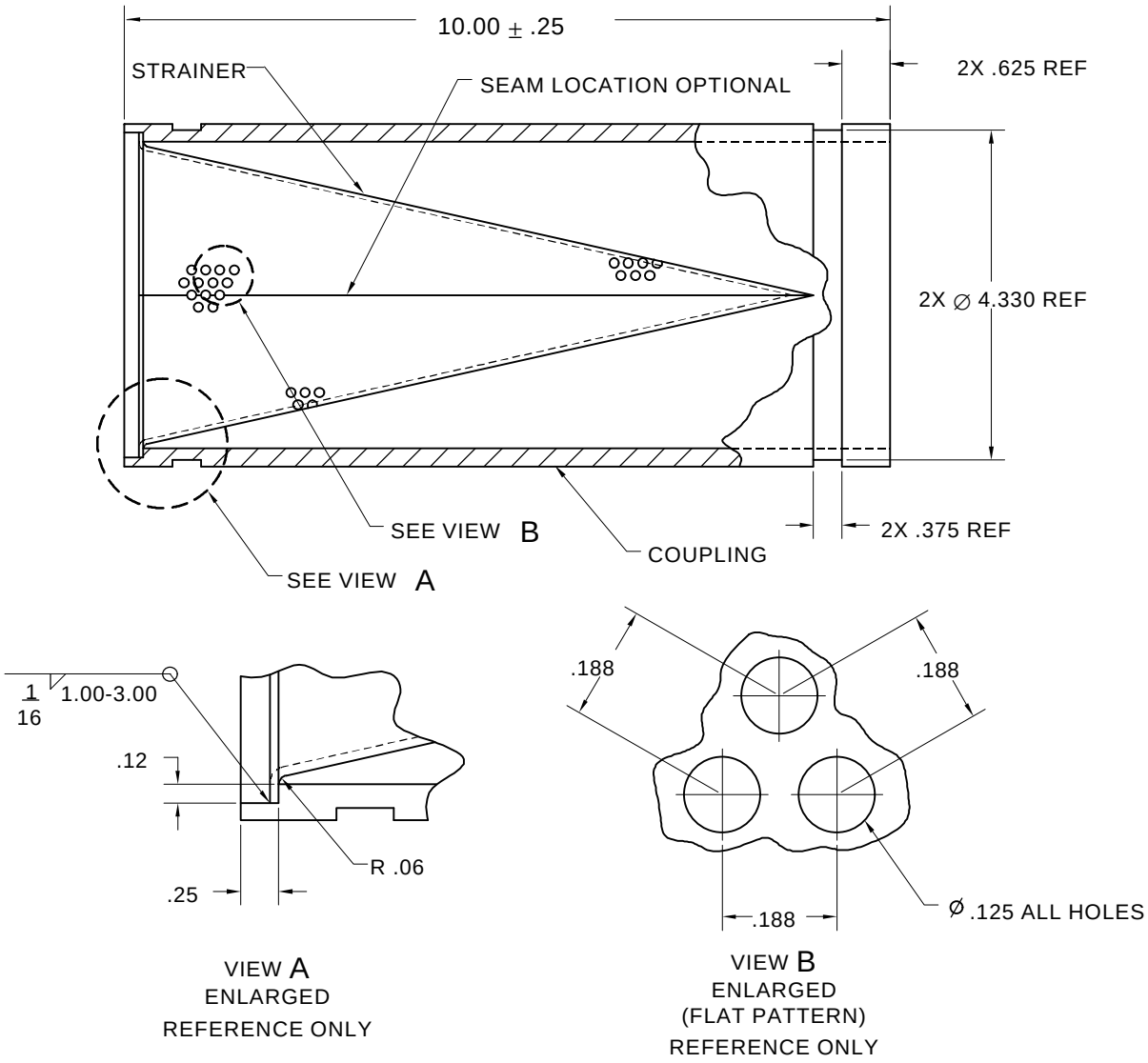
Custodians:
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Review Activity:
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Army – AT

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NOTES:

1. CONFIGURATION SHOWN IS NON-MANDATORY; IT IS ONLY REPRESENTATIVE OF A DESIGN DEEMED ACCEPTABLE
2. DIMENSIONS ARE IN INCHES.
3. UNLESS OTHERWISE SPECIFIED TOLERANCES: .XXX ±.005, .XX ± .03

FIGURE 1. Strainer assembly - Example.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-52345D	2. DOCUMENT DATE (YYMMDD) 990319

3. DOCUMENT TITLE
FLOW CONTROL KIT, HOSELINE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TACOM, AMSTA-TR-D/210 (Fuel and Water Handling)	b. TELEPHONE (Including Area Code) (1) Commercial (810)574-7744 (2) DSN 786-7744
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command Attn: AMSTA-TR-D/210 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 681-9340 DSN 761-9340

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